

General-Relativity

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Lecture 1: Syllabus day

Tue 02 Sep 2025 12:31

We will be using Carroll. Course webpage at <https://physics.bu.edu/~fitzpatr/f25>. Grad class so please call him Liam. Most of the first half of the course is developing the theory of differential geometry, in order to discuss the manifold that is spacetime. Weinberg's book on GR is a good resource as well. Gravity is described by a (not quite) Gauge field $h_{\mu\nu}$ which we will discuss at the end of the course. This implies the equivalence principle which implies gravity behaves as if spacetime is a Riemannian manifold, but it is more fundamental meaning the geometry is not the truth of the universe in the physical sense, only the mathematical sense.

Newtonian gravity is not Lorentz invariant. Fixing this requires the equivalence principle, which states the laws of physics are the same in any inertial reference frame. A great way to phrase this is that standing in a box accelerating upwards is the same as standing in a box that is on Earth, subject to a downwards pull. This means gravity may be described by outwards acceleration.

Interpret gravity as outwards acceleration in this way. This then amounts to sowing together *different* accelerating frames around the border of the Earth. Wow I wonder if this is related to manifolds?

Chapter 1

Review of special relativity

We take the convention that x^μ is a 4-vector with $x^0 = t$, $x^1 = x$, $x^2 = y$, and $x^3 = z$.

Spacetime is obviously a 4-manifold, but it admits a metric, since distance can be measured. The speed of light is the same in all reference frames, so if we fix two coordinate systems x^μ and $(x')^\mu$ for spacetime, they must agree on the speed of light. More precisely, given an initial point x_0^μ and final point x_f^μ for a ray of light, if $\Delta x^0 = x_f^0 - x_0^0$ is the distance in x (and the same for other coordinates), then the time elapsed must be

$$c\Delta t = \sqrt{\Delta x^2 + \Delta y^2 + \Delta z^2}.$$

Equivalently,

$$0 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2.$$

Invoking that the speed of light is the same in these coordinate systems,

$$0 = c^2\Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 = c^2\Delta(t')^2 - \Delta(x')^2 - \Delta(y')^2 - \Delta(z')^2.$$

We define this quantity as the invariant distance between x^μ and $(x')^\mu$. Equivalently,

$$\Delta s^2 = \eta_{\mu\nu} \Delta x^\mu \Delta x^\nu,$$

where

$$\eta_{\mu\nu} = \text{diag}(1, -1, -1, -1)$$

is the Minkowski metric. Here we recall Einstein notation that repeated indicies indicate implicit summations. Since $\eta_{\mu\nu} \neq 0$ iff $\mu = \nu$, the claimed equality follows.

Remark 1.0.1. Carroll uses $x^{\mu'}$ instead of $(x')^\mu$ for different coordinate symbols. This is just stupid because μ' should still be μ ????? Like why would we ever do this wtf

Definition 1.0.2. The manifold $\mathbb{R}^4$ with topology given by Δs^2 is *Minkowski space*. The spacetime metric is also known as the *Minkowski metric*. I will often denote this space by $\mathcal{M}$ for simplicity; this is nonstandard as far as I am aware, but I like naming my mathematical objects.

(Minkowski is pronounced “Minkovski”). When we refer to “flat space” in this course, we always mean Minkowski space. As we do in geometry, we now consider symmetries of flat space. Clearly translations $x^\mu \mapsto (x')^\mu = x^\mu + a^\mu$ for any fixed $a^\mu \in \mathcal{M}$ are metric-invariant homeomorphisms of Minkowski space. However, we are also interested in the Lorentz transformations

Definition 1.0.3. Lorentz transformations Λ_ν^μ are linear maps on x^μ that preserve Δs^2 . They are given by $x^\mu \mapsto \Lambda_\nu^\mu x^\nu$, where the tensor Λ_ν^μ is a 4×4 matrix we describe later.

We equivalently have

$$\Delta x^\mu \Delta x^\nu \eta_{\mu\nu} = \Lambda_\alpha^\mu \Delta x^\alpha \Lambda_\beta^\nu \Delta x^\beta \eta_{\mu\nu}.$$

We thus need

$$\eta_{\alpha\nu} = \Lambda_\alpha^\mu \Lambda_\beta^\nu \eta_{\mu\nu}.$$

Lorentz transformations are any map such that this is true.

Definition 1.0.4. We don't necessarily need to fix Δs^2 , we need to lead the metric for *light rays* invariant, where $\Delta s^2 = 0$. Maps that do this are called *conformal transformations*.

We are using a larger set than usual for Lorentz transformations. Oftentimes linear transformations must be homotopic to the identity, but our definition is more general. *Discrete symmetries* do this. For example, *time parity* $T : x^0 = t \mapsto -t$. Of course T is not homotopic to 1, but it is clearly a metric-invariant homeomorphism. In general, any spacial parity $x^i \mapsto -x^i$ with $i \in \{1, 2, 3\}$ fixed is also a transformation.

Oftentimes we do not want to discuss discrete transformations, so we impose the following requirements:

1. $\Lambda_0^0 \geq 1$,
2. $\det \Lambda = 1$.

The first gets rid of $t \mapsto -t$, and the second kind of just refuses stretching.

Notation 1.0.5. We will often have to deal with more general metrics than just Minkowski's metric. We will generally use the notation $g_{\mu\nu}$, and we define $g^{\mu\nu} \mapsto g_{\mu\nu}^{-1}$, meaning

$$g^{\mu\nu} g_{\nu\rho} = \delta_\rho^\mu = 1,$$

where δ_ρ^μ is the Kronecker delta $\delta_\mu^\nu = 1$ iff $\mu = \nu$, 0 otherwise. Everything else raises and lowers indices with the metric, meaning

$$x^\mu := g^{\mu\nu} x_\nu, \quad x_\mu := g_{\mu\nu} x^\nu.$$

The Minkowski metric is self-inverse $\eta_{\mu\nu} = \eta^{\mu\nu}$. Using this notation, viewing $\eta_{\mu\nu}$ as a vector in μ , we can define

$$\eta_\nu^\mu := \eta^{\mu\alpha} \eta_{\alpha\nu} = 1.$$

For this reason, we will almost never write g_μ^ν or g_ν^μ , since they are both 1.

Lecture 2: Vectors and dual vectors

Thu 04 Sep 2025 12:31

In order to define vectors on a manifold, we work with the *tangent space*. Let M be an n -manifold, and choose some path $x^\mu(-)$ parametrized by the variable λ . Composing along the charts gives some $g : M \rightarrow \mathbb{R}$, and we can compute partial derivatives

$$\frac{\partial g(x)}{\partial x^\mu}$$

by once again composing along the charts in the other direction, which yields some $\mathbb{R}^n \rightarrow \mathbb{R}$. The atlas condition guarantees invariance under choice of charts. The tangent space at $p = x^\mu(\lambda)$, written as $T_p M$, is the $\mathbb{R}$ -vector space of these derivatives. It is spanned by

$$\left\{ \frac{\partial}{\partial x^1}, \dots, \frac{\partial}{\partial x^n} \right\}.$$

Any vector v of this space admits a basis decomposition

$$v = v^\mu \frac{\partial}{\partial x^\mu}.$$

There is a vector

$$\frac{\partial}{\partial x^\mu},$$

with ν th component

$$\left(\frac{\partial}{\partial x^\mu} \right)^\nu = \delta_\mu^\nu \left(\frac{\partial}{\partial x^\mu} \right).$$

Then

$$\frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial x^\nu} \right)^\mu \frac{\partial}{\partial x^\nu}.$$

Oftentimes we write

$$\hat{e}_\mu = \frac{\partial}{\partial x^\mu} = \hat{e}_\mu^\nu \frac{\partial}{\partial x^\nu}$$

for simplicity.

Given two coordinate systems $x^\mu, (x')^\mu$, the tangent space $T_p M$ takes both $\frac{\partial}{\partial x^\mu}$ and $\frac{\partial}{\partial (x')^\mu}$ as bases. To translate between bases, we use the chain rule

$$\frac{\partial}{\partial x^\mu} = \frac{\partial (x')^\alpha}{\partial x^\mu} \frac{\partial}{\partial (x')^\alpha}.$$

We thus obtain a change of basis: given

$$v = v^\mu \frac{\partial}{\partial x^\mu} = (v')^\mu \frac{\partial}{\partial (x')^\mu},$$

we have

$$v = v^\mu \frac{\partial (x')^\alpha}{\partial x^\mu} \frac{\partial}{\partial (x')^\alpha} = (v')^\alpha \frac{\partial}{\partial (x')^\alpha}.$$

Thus,

$$v^\mu \frac{\partial (x')^\alpha}{\partial x^\mu} = (v')^\alpha.$$

We often have functions of vectors. We write $f(v)$ by $f(v^\mu)$, and leave the basis component implicit (i.e. not writing $f(v^1, \dots, v^n)$).

Let $x^\mu(\lambda)$ be a path in M . We define the *total* derivative

$$\frac{dx^\mu(\lambda)}{d\lambda}$$

with components given by the components in the tangent space of $\frac{dx}{d\lambda}$. The total derivative can indeed be computed since paths are functions in 1 variable. Viewed in the tangent space, given

$$V := \frac{dx}{d\lambda},$$

we have

$$V = V^\mu \frac{\partial}{\partial x^\mu}.$$

How does V act on a function g on the manifold?

$$Vg(x) = V^\mu \frac{\partial}{\partial x^\mu} g(x) = \frac{dx^\mu}{d\lambda} \frac{\partial}{\partial x^\mu} g(x) = \frac{d}{d\lambda} g(x(\lambda))$$

where the last equality is precisely the reverse of the chain rule.

We have a metric $g_{\mu\nu}$. Given vectors v^μ, w^ν , we can compute the *inner product* $\langle v, w \rangle = g_{\mu\nu} v^\mu w^\nu$, and we write $g(v, w)$ for this inner product. We generally write this as

$$v^\mu w_\mu, \quad w_\mu := g_{\mu\nu} w^\nu.$$

This agrees with our previous raising-lowering indices convention. We call vectors w_μ *dual vectors*. Dual vectors have a *dual basis* dx^μ given by

$$dx^\mu \left(\frac{\partial}{\partial x^\nu} \right) = \delta^\mu_\nu.$$

Dual vectors essentially act on vectors, rather than on functions. A general dual vector ω admits a basis decomposition via

$$\omega = \omega_\mu dx^\mu.$$

Linearity thus extends the formula for dual vectors acting on vectors:

$$\omega(v) = \omega_\mu dx^\mu \left(v^\nu \frac{\partial}{\partial x^\nu} \right) = x_\mu v^\nu dx^\mu \frac{\partial}{\partial x^\nu} = \omega_\mu v^\nu \delta^\nu_\mu = \omega_\mu v^\mu.$$

Consider transformations under basis change. Choose $x^\mu, (x')^\mu$ coordinates. Since $\omega(v) = \omega_\mu v^\mu$ is basis independent,

$$\begin{aligned} \omega_\mu v^\mu = \omega'_\alpha (v')^\alpha &\implies \omega'_\alpha = \omega_\nu \frac{\partial x^\nu}{\partial (x')^\alpha} \implies \omega'_\alpha (v')^\alpha = \omega_\nu \frac{\partial x^\nu}{\partial (x')^\alpha} \left(v'^\beta \frac{\partial (x')^\alpha}{\partial x^\beta} \right) \\ &= v'_\nu \delta^\nu_\alpha = \omega_\nu v^\nu. \end{aligned}$$

We finally are able to rigorize differentials. Let $\varphi(x)$ a function of M , and recall we define

$$d\varphi = \frac{\partial \varphi(x)}{\partial x^\mu} dx^\mu.$$

Some books write

$$\varphi, \mu = \frac{\partial \varphi}{\partial x^\mu}.$$

Question: are “dual vectors” actually elements (at least, up to isomorphism) of the dual vector space?

Lecture 3: Locally inertial coordinates

Tue 09 Sep 2025 12:29

We will now begin discussing tensors. We begin with the metric tensor $g_{\mu\nu}$. Recall that components of the dual vector transform by

$$\omega_\mu \rightarrow \omega'_\mu = \frac{\partial x^\alpha}{\partial (x')^\mu} \omega_\alpha.$$

The metric tensor transforms the same way in 2 indices, meaning

$$g_{\mu\nu} \rightarrow g'_{\mu\nu} = \frac{\partial x^\alpha}{\partial (x')^\mu} \frac{\partial x^\beta}{\partial (x')^\nu} g_{\alpha\beta}.$$

This implies there is some basis independent quantity g , and the components $g_{\mu\nu}$ are a choice in a basis, meaning

$$g = g_{\mu\nu} dx^\mu dx^\nu = g'_{\mu\nu} d(x')^\mu d(x')^\nu.$$

This means the metric's components are not invariant under coordinate change. This is useful: let M be spacetime, and choose a point x_p^μ in M . WLOG, there exist coordinates such that $g_{\mu\nu}(x_p^\mu) = \eta_{\mu\nu}$ and such that the partials are 0:

$$\partial_p g_{\mu\nu}(x_p^\mu) = 0,$$

where we have introduced the very convenient notation

$$\partial_\mu = \frac{\partial}{\partial x^\mu}.$$

We call such coordinates *locally inertial coordinates*.

Let x^α and $(x')^\mu$ be coordinate systems. Fix x_p^μ a point, and compute the Taylor expansion $(x')^\mu$ about x_p^μ :

$$(x')^\mu(x) = (x')^\mu(x_p) + (x - x_p)^\alpha (\partial_\alpha (x')^\mu(x_p)) + \frac{1}{2} (x - x_p)^\alpha (x - x_p)^\beta (\partial_\alpha \partial_\beta (x')^\mu_p) + \dots$$

The first few terms give us the fact that $g_{\mu\nu}$ has $\frac{d(d+1)}{2}$ parameters, but we may choose d^2 total free parameters. Thus, we may choose $g_{\mu\nu} = \eta_{\mu\nu}$ and still have $\frac{d(d-1)}{2}$ free parameters left over. Physically, these free parameters that leave the Minkowski metric invariant are the Lorentz transformations: velocity boosts and rotations. We think of these Lorentz transformations as “generating” elements: a rotation in $\mathbb{R}^2$ for example should be a “generating element” (interpreted as an infinitesimal) for $SO(2)$.

We will almost always assume locally inertial coordinates. We will choose a point p , and assume WLOG that the first derivatives vanish at that point and the metric at that point is Minkowski. This is useful because we know the laws of physics in Minkowski space.

Chapter 2

Tensors

We will now introduce tensors in a differential geometric context. We have already seen one: $g = g_{\mu\nu} dx^\mu \otimes dx^\nu$. Earlier, we didn't write the tensor product, but now we do. This is an element of the tensor square of the dual space $(T_p^* M)^\otimes 2$. Similarly, $g^{-1} = g^{\mu\nu} \partial_\mu \otimes \partial_\nu$ is an element of the tensor product of the tangent space. We may also write $1 = \delta_\mu^\nu \partial_\mu \otimes dx^\nu$. These are called $(0, 2)$ -, $(2, 0)$ -, and $(1, 1)$ -tensors, respectively. We will never again write the tensor product and will leave it implicit.

Of course, we may generalize this to (r, s) -tensors which are given as

$$T = T_{\nu_1 \dots \nu_s}^{\mu_1 \dots \mu_r} \partial_{\mu_1} \dots \partial_{\mu_s} dx^{\nu_1} \dots dx^{\nu_s}.$$

These of course may be interpreted as functions on the relevant space. Matrices may be viewed as bilinear via the identification $M(v_1, v_2) := M_{ij} v_1^i v_2^j$. (r, s) -tensors generalize this in the sense that they are (r, s) -linear functions: they are r -multilinear in vectors and s -multilinear in dual vectors by acting on vectors in precisely the same way. We can always use the metric to raise and lower indices, meaning we can obtain a $(0, r + s)$ -tensor (or likewise in the tangent space tensor power) by lowering indices on an (r, s) -tensor:

$$T_{\mu_1 \dots \mu_r \nu_1 \dots \nu_s} = g_{\mu_1 \alpha_1} \dots g_{\mu_r \alpha_r} T_{\nu_1 \dots \nu_s}^{\alpha_1 \dots \alpha_r}.$$

We may also be interested in making a tensor T into a symmetric tensor, wherein $T_{ij} = T_{ji}$ (without writing extra indices), or antisymmetrizing it by $T_{ij} = -T_{ji}$. We can do this with matrices: given M_{ij} a matrix,

$$\frac{M_{ij} + M_{ji}}{2}$$

is symmetric, and subtracting gives an antisymmetric matrix. We do a similar process with matrices: given σ any permutation on $\{1, \dots, n\}$, we write

$$T_{(\mu_1 \dots \mu_n)} := \frac{1}{n!} \sum_{\sigma} T_{\sigma}.$$

We only symmetrize on things in the parentheses, meaning

$$T_{(\mu\nu)\rho} = \frac{1}{2} (T_{\mu\nu\rho} + T_{\nu\mu\rho})$$

only symmetrizes μ, ν . Similarly, $T_{(\mu|\nu|\rho)}$ means we symmetrize μ and ρ ; the line is a “break”. Notation for antisymmetrizing is given by brackets

$$T_{[\mu\nu]} = \frac{1}{2} (T_{\mu\nu} - T_{\nu\mu}).$$

These are important due to their relation to Lorentz invariance. For example, we apply this to Maxwell's equations. We write these componentwise as

$$\begin{aligned}\varepsilon^{ijk}\partial_j B_k - \partial_0 E^i &= J_E^i, \\ \partial_i E^i - J_E^0 &, \\ \varepsilon^{ijk}\partial_j E_k + \partial_0 B^i &= J_M^i, \\ \partial_i B^i &= -J_m^0.\end{aligned}$$

Lecture 4: idk

Thu 11 Sep 2025 12:36

Missed some stuff, read notes.

The energy-momentum tensor or stress tensor $T_{\mu\nu}$ is how energy and momentum enter into the equations of motion, since the stress tensor turns energy and momentum and turns them into an object which transforms like a tensor.

A simple example is the case of a fluid. We can view this as a collection of individual molecules, and (in theory) we could keep track of the trajectory of each particle as some $x^\mu(\lambda) = (t(\lambda), x^i(\lambda))$. There are many ways to parameterize x^μ , so we choose to parameterize it with the *proper time* τ . This is the amount of time that elapses for the *particle itself*: if it was carrying a stopwatch, it's the readout of the stopwatch. Using Δs , we find

$$\left(\frac{d\tau}{d\lambda}\right)^2 = -\eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}.$$

This basically puts the particle in its rest frame. In the rest frame,

$$\frac{dx^i}{d\lambda} = 0$$

since the particle is not moving. Therefore,

$$0 = -\eta_{\mu\nu} dx^\mu dx^\nu = -\eta_{00} dx^0 dx^0 = dt^2.$$

Convention: particle physicists take $(+, -, -, -)$ for $\eta_{\mu\nu}$, and relativists take the opposite convention $(-, +, +, +)$. We are going to swap to the relativistic conventions now. This is why there is a $-\eta_{\mu\nu}$ in the equation.

Therefore, in the rest frame,

$$d\tau^2 = dt^2 \implies \left(\frac{d\tau}{d\lambda}\right)^2 = \left(\frac{dt}{d\lambda}\right)^2.$$

We can also view fluids macroscopically. The quantity

$$u^\mu := \frac{dx^\mu}{d\tau}$$

can be viewed as a continuous function of spacetime by viewing it as an average over the little x^μ s. We then have

$$u^\mu u_\mu = \eta_{\mu\nu} u^\mu u^\nu = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = \eta_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{d\tau}{d\lambda} \frac{dx^\nu}{d\lambda} \frac{d\tau}{d\lambda}.$$

Recall of course that in the rest frame, the space components go to 0 and τ is proper time, so plugging in our previous equations we find it normalizes to -1 .

Let's find $T_{\mu\nu}$ for a perfect fluid. The stress tensor locally depends on u^μ , and in the local rest frame its rotationally invariant. We first look at $T^{\mu\nu}$ in the rest frame. This is a symmetric 4x4 matrix. For it to be rotationally invariant, all non-diagonal elements have to vanish. This can be proven by some annoying math. We also find that due to $\eta_{\mu\nu}$ signs, the first term $\rho = T^{00}$ is energy density, and the other terms T^{ii} are pressure. This follows because space and time are different:

In a perfect fluid, x, y, z are all the same, so rotations in 3d don't do anything, and only 1 is invariant under 3d rotations. We are not demanding that we do 4-dimensional rotations (boosts), since it is not invariant under these. This gives us a second degree of freedom in ρ .

Aside: If we required it be invariant under boosts, we would have $\rho = -p$ due to the metric. This would imply that for a Lorentz transformation Λ ,

$$T^{\mu\nu} = \Lambda \eta^{\mu\nu} = \begin{pmatrix} -\Lambda & & & \\ & \Lambda & & \\ & & \Lambda & \\ & & & \Lambda \end{pmatrix}.$$

This is dark energy apparently.

We have simplified $T^{\mu\nu}$ to

$$T^{\mu\nu} = \text{diag}(\rho, p, p, p).$$

We then lower indices

$$T_{\mu\nu} = \eta_{\mu\alpha} \eta_{\nu\beta} T^{\alpha\beta}.$$

Useful trick. We start by deriving a formula in a specific frame, and we then use tensor analysis to write the tensor in any frame. We will now do this with $T_{\mu\nu}$, since we know it in its rest frame. We start by writing $T^{\mu\nu}$ in terms of $u^\mu = (1, 0, 0, 0)$ in the rest frame. We of course have $u^\mu u^\nu = \text{diag}(1, 0, 0, 0)$, so we can start with $\rho u^\mu u^\nu$.

The trick requires us to only use the 2-tensor $\mu\nu$, so writing it down we have to use the two indices only. Thus, the things we can put on the right-hand side are $u^\mu u^\nu$ and $\eta^{\mu\nu}$. In other words, the fluid picked out a specific reference frame, so we can only use things which pick out the same reference frame, which are u and the spacetime metric, since the latter is invariant under reference frames. We have

$$p\eta^{\mu\nu} + (\rho + p)u^\mu u^\nu = T^{\mu\nu}.$$

This gives us a way to express $T^{\mu\nu}$ in a general frame, by invariance under coordinate transformations. We simply have

$$T^{\mu\nu} = p\eta^{\mu\nu} + (\rho + p)u^\mu u^\nu$$

in *any* frame. Finally, note that energy and momentum are conserved, meaning we must have

$$\partial_\mu T^{\mu\nu} = 0.$$

This follows for this example without extra work I think.

Chapter 3

Equivalence principle

Our previous discussion forms the basis for the equivalence principle, which is so important that it is basically a definition.

Definition 3.0.1. The *weak equivalence principle* states that forces on particles only depend on distance:

$$m\mathbf{a}_i = \mathbf{F}_i = -m\Delta\Phi_i = \sum_{j \neq i} \mathbf{F}(\mathbf{x}_i - \mathbf{x}_j).$$

We can thus always remove gravity by passing to an accelerating frame: given the old coordinate system x^μ , we map to a new system $(x')^\mu$ by

$$x^\mu \rightarrow x^\mu + \frac{1}{2}t^2\Delta\Phi.$$

The *strong* equivalence principle states that all laws of physics are exactly the same in any (inertial) reference frame, and are invariant under coordinate transformations.

Mass comes from interactions. The mass of the proton, for example, is 1GeV, but the mass of quarks are only a few MeV. The mass of the proton comes largely from the interactions of its quarks, so forces have to couple to these interactions such that the Newtonian weak equivalence principle is satisfied. Thus, mass is inherently relativistic, and it's hard for the strong equivalence principle to not be true.

For example, effects of gravity are precisely effects of non-gravitational acceleration. This allows us to convert problems of gravity to problems that we already know: acceleration without gravity. Let's say one twin is standing on Earth and one twin is *accelerating* in a rocket. The twin that is standing still is normal, but the moving twin experiences time dilation. If both are in inertial frames, nothing weird happens, but if the other accelerates to come back, the one who was moving is younger. This tells us that if we switch this scenario to being in a gravitational field, we find that we age slower in gravitational fields.

Lecture 5: Covariance

Tue 16 Sep 2025 12:32

Before discussing covariance, we continue our discussion of the Equivalence principle. Consider a laser firing a photon at a fixed wavelength λ_i . We measure it at the top with a detector to be a *different number* $\lambda_f \neq \lambda_i$. This experiment was done at Harvard by shooting photons 22 meters. This small difference in gravity was able to measure the redshift. We derive this using only the equivalence principle.

This situation is equivalent to the entire tower accelerating upwards at $9.8m/s^2$, and shooting a laser up the tower the instant it starts accelerating. The distance $z(t)$ of the photon from z_0 is $z(t) = z_0 + \frac{1}{2}at^2$. Of course, $\Delta t = t_f - t_i = \frac{z_f}{c}$. We choose $t_i = 0$. The velocity of the rocket is simply $v = at_f = a\frac{z_0}{c}$. There is then a redshift factor of $\frac{\Delta\lambda}{\lambda_i}$ by the Doppler effect. This gives a doppler shift of

$$\sqrt{\frac{1 + v/c}{1 - v/c}} - 1.$$

This approximates out to az_f/c^2 in the small gravitational potential when plugging in the ts .

We use a mathematical systemization of this argument, called *general covariance*.

1. First, write down equations of motion for physics in the absense of gravity.
2. Then, introduce the metric and its derivatives into the equations of motion in such a way that the resultant equations of motion are invariant under coordinate changes (Lorentz transformations). We then WLOG reduce to Minkowski space.

To do this, we need to learn how to efficiently write the equations of motion in a coordinate-invariant manner more efficiently. We will explore this at length. The fascinating implications of this fact are that gravity follows completely by coordinate-invariance and the Minkowski metric, meaning the *entire effect of gravity* is the coordinate transformation into the gravitational frame.

We start by considering free motion of a particle

$$\frac{du^\mu}{d\tau} = 0.$$

Recall that u^μ is the 4-velocity (u^0, u^i) and τ is the proper time (rest frame time)

$$d\tau^2 = -\eta_{\mu\nu}dx^\mu dx^\nu.$$

. Idk τ represents time in our coordinate system. This is equivalently written as

$$\frac{d^2x^\mu}{d\tau^2} = \left(\frac{d^2t}{d\tau^2}, \frac{d^2x^i}{d\tau^2} \right) = 0.$$

In the rest frame, $t = \tau$, so the time component vanishes in the rest frame.

The issue is we want to also work with massless particles, and they don't have proper time. This means we can't use τ since it never changes, so we have to do something else. Define $\sigma = x^0$ as the time component in an inertial frame that is not necessarily the reference frame. We then have

$$\frac{d^2x^\mu}{d\sigma^2} = 0.$$

We use affine parameters to show that σ is invariant under our choice. Write $\sigma = a\lambda + b$, where λ is an affine parameter. We then still have

$$\frac{d^2x^\mu}{d\lambda^2} = 0$$

by change of variables. We thus have the freedom to redefine the coordinate system using any affine parameter.

Both massive *and* massless particles can then be written as

$$\frac{d^2\xi^\mu}{d\lambda^2} = 0, \quad \xi^\mu = \text{freely falling frame.}$$

Of course by Lorentz invariance (and in particular, invariance under boosts), we see that the velocity of the particle in this frame need not be defined, since all are Lorentz invariant.

The question is: what are the equations of motion in the lab frame? The answer is given by transforming

$$\frac{d^2\xi^\mu}{d\lambda^2} = 0 \rightarrow \text{lab frame equivalent.}$$

We do this explicitly by computing $\xi^\mu(x)$. We want to write these in terms of the metric. In the freely falling frame we choose the metric by $\eta_{\mu\nu}$ with vanishing first derivatives. This is what the freely falling frame is (recall it exists by earlier discussion). The lab frame is an accelerating frame because our lab is on a rocketship. I love GR.

Write $g_{\mu\nu}$ for the metric for the lab frame, and recall that we have that our metric transformation is given by

$$g_{\mu\nu} \rightarrow g'_{\mu\nu} = g_{\alpha\beta} \frac{\partial x^\alpha}{\partial (x')^\mu} \frac{\partial x^\beta}{\partial (x')^\nu} = \partial_\mu x^\alpha \partial_\nu x'^\beta = g_{\alpha\beta} \frac{\partial x^\alpha}{\partial \xi^\mu} \frac{\partial x^\beta}{\partial \xi^\nu}.$$

A trick to remembering this is that the indices must always contract meaning since μ, ν are on the bottom for $g_{\mu\nu}$, they must also be on the bottom, which we have by $\partial_\mu, \partial_\nu$.

Since g' is the freely falling frame, we have

$$g_{\alpha\beta} \frac{\partial x^\alpha}{\partial \xi^\mu} \frac{\partial x^\beta}{\partial \xi^\nu} = \eta_{\mu\nu}.$$

Writing the tensors as matrices, we have inverse matrices

$$\frac{\partial x^\alpha}{\partial \xi^\mu} = \left(\frac{\partial \xi^\alpha}{\partial x^\mu} \right)^{-1}.$$

Multiplying through on our above equation gives

$$\eta \frac{\partial \xi^\mu}{\partial x^\sigma} = g_{\alpha\beta} \frac{\partial x^\alpha}{\partial \xi^\mu} \frac{\partial \xi^\mu}{\partial x^\sigma} \frac{\partial x^\beta}{\partial \xi^\nu} = g_{\alpha\beta} \delta_\sigma^\alpha \frac{\partial x^\beta}{\partial \xi^\nu} = g_{\sigma\beta} \frac{\partial x^\beta}{\partial \xi^\nu}.$$

Applying the same argument again gives

$$\eta_{\mu\nu} \frac{\partial \xi^\mu}{\partial x^\sigma} \frac{\partial \xi^\nu}{\partial x^\rho} = g_{\sigma\rho}.$$

We are now writing τ instead of λ because the prof's notation got lazy. Applying the chain rule and product rule,

$$0 = \frac{d^2\xi^\mu}{d\tau^2} = \frac{d}{d\tau} \frac{d}{d\tau} \xi^\mu = \frac{d}{d\tau} \left(\frac{dx^\alpha}{d\tau} \frac{\partial \xi^\mu}{\partial x^\alpha} \right) = \frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} + \frac{d^2 x^\alpha}{d\tau^2} \frac{\partial \xi^\mu}{\partial x^\alpha}.$$

Of course throughout this we are assuming derivatives freely commute, which indeed follows for smooth maps. We use the same process to strip off the factor of $\partial_\alpha \xi^\mu$ by multiplying through by $\partial_{\xi^\mu} x^\sigma$:

$$0 = \frac{dx^\sigma}{d\tau^2} + \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \frac{\partial x^\sigma}{\partial \xi^\mu} \frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta}.$$

We write

$$\Gamma_{\alpha\beta}^\sigma := \frac{\partial^2 \xi^\mu}{\partial x^\alpha \partial x^\beta} \frac{\partial x^\sigma}{\partial \xi^\mu}.$$

The answer is then rewritten as

$$0 = \frac{d^2 x^\sigma}{d\tau^2} + \Gamma_{\alpha\beta}^\sigma \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau}.$$

This is called the *geodesic equation*. We want to rewrite it in terms of a metric to be able to use it more easily. To do this, we have to rewrite $\Gamma_{\alpha\beta}^\sigma$ in terms of the metric

$$\eta_{\mu\nu} \frac{\partial \xi^\mu}{\partial x^\sigma} \frac{\partial \xi^\nu}{\partial x^\rho} = g_{\sigma\rho}.$$

Differentiate with respect to x^λ :

$$\frac{\partial g_{\alpha\beta}}{\partial x^\lambda} = \eta_{\mu\nu} \frac{\partial^2 \xi^\mu}{\partial x^\lambda \partial x^\alpha} \frac{\partial \xi^\nu}{\partial x^\beta} + \eta_{\mu\nu} \frac{\partial \xi^\mu}{\partial x^\alpha} \frac{\partial^2 \xi^\nu}{\partial x^\lambda \partial x^\beta}.$$

Notice that this is simply

$$\frac{\partial g_{\mu\nu}}{\partial x^\lambda} = \eta_{\mu\nu} \left(\Gamma_{\lambda\alpha}^\rho \frac{\partial \xi^\mu}{\partial x^\rho} \frac{\partial \xi^\nu}{\partial x^\beta} + \Gamma_{\lambda\beta}^\rho \frac{\partial \xi^\mu}{\partial x^\alpha} \frac{\partial \xi^\nu}{\partial x^\rho} \right) = \Gamma_{\lambda\alpha}^\rho g_{\rho\beta} + \Gamma_{\lambda\beta}^\rho g_{\alpha\rho},$$

where the final step follows also by our above formula for $g_{\mu\nu}$.

Note that by definition, the Christoffel symbol $\Gamma_{\alpha\beta}^\mu$ is symmetric under the lower indices, so we add the equation to itself under a permutation of indices, which I don't want to write. We end up with

$$\partial_\lambda g_{\mu\nu} + \partial_\mu g_{\lambda\nu} - \partial_\nu g_{\mu\lambda} = 2\Gamma_{\lambda\mu}^\rho g_{\rho\nu}.$$

Stripping the metric by multiplying by its inverse:

$$\Gamma_{\alpha\beta}^\rho = \frac{1}{2} g^{\sigma\rho} \left(\frac{\partial g_{\alpha\sigma}}{\partial x^\beta} + \frac{\partial g_{\beta\sigma}}{\partial x^\alpha} - \frac{\partial g_{\alpha\beta}}{\partial x^\sigma} \right).$$

This allows us to rewrite the geodesic equation in terms of the metric only. Since the Minkowski metric has fully dropped out, this holds in *any metric*.

Lecture 6: Covariant derivatives

Thu 18 Sep 2025 12:33

Furthering the topic of covariance, we introduce covariant derivatives. Tensors are automatically invariant under coordinate transformations, but derivatives are not usually tensors. We define a covariant derivative ∇_μ to be a derivative that transforms in the right way. Since ∂_μ transforms properly on scalar functions, we should have $\nabla_\mu f = \partial_\mu f$. We should also have that if S, T are tensors, then $\nabla_\mu(S + T) = \nabla_\mu S + \nabla_\mu T$ (linearity). Finally, it should satisfy the product rule with respect to the tensor product.

Let's see how this acts on a dual vector ω_ν . We should have a tensor $\nabla_\mu \omega_\nu = X_{\mu\nu}$. The only linear maps on dual vectors are derivatives ∂_μ and matrices $C_{\mu\nu}^\alpha$. The most general thing we can have is thus

$$\nabla_\mu \omega_\nu = a \partial_\mu \omega_\nu - C_{\mu\nu}^\alpha \omega_\alpha,$$

for some scalar a . The reason we use a minus sign is so the right side can later become $\Gamma_{\mu\nu}^\alpha$. Condition 1 forces $a = 1$ for scalars, and the product rule forces $a = 1$ for vectors.

Now we consider it on an arbitrary vector v^α . We must have

$$\nabla_\mu v^\alpha = \partial_\mu v^\alpha + B_{\mu\beta}^\alpha v^\beta.$$

We should find a relation between B and C . Interestingly, we can already do this. Since $\omega_\alpha v^\alpha$ is a scalar,

$$\nabla_\mu(\omega_\alpha v^\alpha) = \partial_\mu(\omega_\alpha v^\alpha) = \omega_\alpha \partial_\mu v^\alpha + (\partial_\mu \omega_\alpha) v^\alpha.$$

Alternatively, via the product rule,

$$\nabla_\mu(\omega_\alpha v^\alpha) = (\nabla_\mu \omega_\alpha) v^\alpha + \omega_\alpha \nabla_\mu v^\alpha = \left(\partial_\mu \omega_\alpha - C_{\mu\alpha}^\beta \omega_\beta \right) v^\alpha + \omega_\alpha \left(\partial_\mu v^\alpha - B_{\mu\beta}^\alpha v^\beta \right).$$

These two expressions must be equal. We cancel the partials to obtain

$$0 = -C_{\mu\alpha}^\beta \omega_\beta v^\alpha + B_{\mu\beta}^\alpha \omega_\alpha v^\beta.$$

Intuitively, $B_{\mu\alpha}^\beta = -C_{\mu\alpha}^\beta$. To prove this, since α, β are dummy indices, we can rename them as

$$C_{\mu\alpha}^\beta \omega_\beta v^\alpha = B_{\mu\alpha}^\beta \omega_\beta v^\alpha.$$

Since this holds for *every* dual vector and *every* vector, this gives our relation. This can be done pointwise by choosing ω, v to pick out individual entries. We now have

$$\nabla_\mu v^\alpha = \partial_\mu v^\alpha + C_{\mu\beta}^\alpha v^\beta.$$

The product rule can generalize to all tensors by taking the product over vectors and duals. Arbitrary tensors generalize by taking the derivative, then adding a term for each index, viewing upper indices as like vectors and lowers like duals.

We are basically correcting the derivative to make it transform like a tensor by adding all the other terms. Any tensor satisfying these properties is a covariant derivative. We specialize by choosing a nice covariant derivative for GR.

We want $C_{\mu\nu}^\rho = C_{\nu\mu}^\rho$, meaning C is symmetric in μ, ν . This is nice for physics reasons, since we can just symmetrize it and antisymmetrize it. The antisymmetrization transforms like a tensor, and if that existed, then that would be an additional field. We have not found this extra field, so we demand it not have an antisymmetric component.

We also require that $\nabla_\mu g_{\alpha\beta} = 0$. We call this metric compatibility. This will help develop the equations of motion from the equivalence principle. This fixes ∇_μ as follows. We have

$$\nabla_\mu g_{\alpha\beta} = \partial_\mu g_{\alpha\beta} - C_{\mu\alpha}^\lambda g_{\lambda\beta} - C_{\mu\beta}^\lambda g_{\alpha\lambda}.$$

We have

$$\partial_\mu g_{\alpha\beta} = C_{\mu\alpha}^\lambda g_{\lambda\beta} + C_{\mu\beta}^\lambda g_{\alpha\lambda}.$$

Since the partial of the metric is $\partial_\mu g_{\alpha\beta} = \Gamma_{\mu\alpha}^\lambda g_{\lambda\beta} + \Gamma_{\mu\beta}^\lambda g_{\alpha\lambda}$, we just obtain

$$C_{\alpha\beta}^\lambda = \Gamma_{\alpha\beta}^\lambda.$$

Here is an example. The metric for flat space in polar coordinates is

$$ds^2 = -dt^2 + dr^2 + r^2 d\theta^2.$$

We thus have $g_{\mu\nu} = \text{diag}(-1, 1, r^2)$. Computing $\Gamma_{\theta\theta}^r$ at a point:

$$\Gamma_{\theta\theta}^r = \frac{1}{2} g^{r\rho} (2\partial_\theta g_{\rho\theta} - \partial_\rho g_{\theta\theta}) = \frac{1}{2} g^{rr} (2\partial_\theta g_{r\theta} - \partial_r g_{\theta\theta}) = \frac{1}{2} \frac{1}{r^2} (-\partial_r r^2) = -r.$$

Lecture 7: Geodesics

Tue 23 Sep 2025 12:30

Consider a vector v^μ tangent to a path $x^\mu(\lambda)$. We have

$$\frac{D}{d\lambda} := \frac{dx^\mu}{d\lambda} \nabla_\mu.$$

Applying this to v gives

$$0 = \frac{D}{d\lambda} v^\mu = \frac{dx^\nu}{d\lambda} (\partial_\nu v^\mu + \Gamma_{\nu\alpha}^\mu v^\alpha).$$

I skipped something. We want to check as claimed that if two vectors v^μ, w^μ on a path $x^\mu(\lambda)$, then when parallel transported along the path, the angle doesn't change. The $D/d\lambda = 0$ is parallel transport or something.

$$\frac{D}{d\lambda} (v^\mu w_\mu) = 0.$$

If they are parallel transported, then

$$\frac{D}{d\lambda} v^\mu = \frac{D}{d\lambda} w^\mu = 0.$$

Computing:

$$\frac{D}{d\lambda} (v^\mu w_\mu) = \frac{dx^\nu}{d\lambda} \nabla_\nu (v^\mu w_\mu).$$

By the product rule, we see

$$= \frac{dx^\nu}{d\lambda} (\nabla_\nu (v^\mu) w_\mu + v^\mu \nabla_\nu w_\mu).$$

The first term goes to 0 by hypothesis, and the second term has its index raised by the metric

$$= \frac{dx^\nu}{d\lambda} (v^\mu \nabla_\nu g_{\mu\alpha} v^\alpha) = \frac{dx^\nu}{d\lambda} v^\mu (g_{\mu\alpha} \nabla_\nu v^\alpha + g_{\mu\alpha} \nabla_\nu v^\alpha).$$

The second term is 0 by hypothesis, and the first is 0 since the covariant derivative of the metric vanishes by construction.

What happens if we require $\frac{dx^\mu}{d\lambda}$ be parallel transported, meaning the angle we are traveling in is invariant along the curve? Then

$$0 = \frac{D}{d\lambda} \left(\frac{dx^\mu}{d\lambda} \right) = \frac{dx^\nu}{d\lambda} \nabla_\nu \left(\frac{dx^\mu}{d\lambda} \right) = \frac{dx^\nu}{d\lambda} \left(\frac{\partial}{\partial x^\nu} \frac{dx^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu \frac{dx^\sigma}{d\lambda} \right) = \frac{d}{d\lambda} \frac{dx^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda}.$$

The last equality in the first term is chain rule. This is simply the geodesic equation:

$$0 = \frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda}.$$

Thus, the geodesic equation is simply the requirement that the derivative be parallel transported.

We would like to simplify computation of Christoffel symbols. Recall that given a distance dx^μ , the change in proper time is

$$d\tau^2 = -g_{\mu\nu} dx^\mu dx^\nu = -g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} d\lambda^2.$$

We write

$$\tau = \int d\tau = \int \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}} d\lambda.$$

We are going to extremize τ by taking the variance $\delta\tau$ with respect to x^μ :

$$\delta\tau = \int \frac{1}{2} \left(-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{-1/2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{d\delta x^\nu}{d\lambda} \right) d\lambda.$$

Of course, $\delta x^\mu = \delta(x^\mu)$. Since μ, ν are contracted with the metric, they are dummy variables, so we can rename them

$$\delta\tau = \int \frac{1}{2} \left(-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{-1/2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - g_{\mu\nu} \frac{dx^\nu}{d\lambda} \frac{d\delta x^\mu}{d\lambda} \right) d\lambda.$$

Of course, the last two terms are now the same, so since derivatives commute,

$$\delta\tau = \int \frac{1}{2} \left(-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{-1/2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - 2g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) d\lambda.$$

Note that

$$\left(-g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right)^{-1/2} = \left(\frac{d\tau^2}{d\lambda^2} \right)^{-1/2} = \frac{d\lambda}{d\tau}.$$

We then have

$$\begin{aligned} \delta\tau &= \int \frac{1}{2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - 2g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) \frac{d\lambda}{d\tau} d\lambda \\ &= \delta\tau = \int \frac{1}{2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} - 2g_{\mu\nu} \frac{d\delta x^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} \right) \left(\frac{d\lambda}{d\tau} \right)^2 d\tau. \end{aligned}$$

We have

$$\frac{\partial x^\mu}{\partial \lambda} = \frac{\partial x^\mu}{\partial \tau} \frac{\partial \tau}{\partial \lambda},$$

giving

$$\delta\tau = \int \frac{1}{2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \left(\frac{d\tau}{d\lambda} \right)^2 - 2g_{\mu\nu} \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau} \left(\frac{d\tau}{d\lambda} \right)^2 \right) \left(\frac{d\lambda}{d\tau} \right)^2 d\tau.$$

The derivatives cancel, yielding

$$\delta\tau = \int \frac{1}{2} \left(\left(-\frac{\partial}{\partial x^\sigma} g_{\mu\nu} \right) \delta x^\sigma \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} - 2g_{\mu\nu} \frac{d\delta x^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right) d\tau.$$

Apparently we can just write this as

$$\delta \left(\frac{1}{2} \int -g_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} d\tau \right).$$

This is a nicer variational action to use, since the square root is gone. Interestingly, the integrand is 1 by a previous equation. This is not quite the version we want to use, since we have a derivative of the variation. We want it to just be an integral times dx^μ . The EOM is then integrand equal to 0.

To fix this, we do some integration by parts, and the boundary vanishes since there is no variation at the boundary.

Remark 3.0.2. the δ is just a derivative. but with... different symbols.

I don't know why we're going on the tangent we can just recall the geodesic equation and definition of the Christoffel symbol in terms of the metric:

$$\delta\tau = \int g_{\sigma\rho} \left(\frac{d^2x^\rho}{d\tau^2} + \Gamma_{\alpha\beta}^\rho \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} \right) \delta x^\sigma d\tau.$$

The extremization is given by just taking the integrand and dropping the δx^μ by the fundamental theorem of the calculus of variations. Wow, the equation of motion is just the geodesic equation.

If $g_{\mu\nu}$ is arbitrary, then Christoffel symbol computations are just brute force. However, if the metric has symmetries (eg. is symmetric), then plugging into the action principal can simplify computations.

So

$$0 = 2g_{\sigma\beta} \left(\frac{d^2\beta}{d\tau^2} + \Gamma_{\mu\nu}^\beta \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \right).$$

Let's do an example with flat space in polar coordinates $ds^2 = -dt^2 + dr^2 + r^2d\theta^2$.

$$L \equiv \frac{1}{2} \int -g_{\alpha\beta} \frac{dx^\alpha}{d\tau} \frac{dx^\beta}{d\tau} d\tau = \frac{1}{2} \int \left(\frac{dt^2}{d\tau} - \frac{dr^2}{d\tau} - \frac{dr^2}{d\tau} \frac{d\theta^2}{d\tau} \right) d\tau.$$